

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	20 July 2026
Team ID	
Project Name	Plugging Into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion & Cleaning	Import raw dataset into SQL Server Handle missing values and duplicates Standardize state/region names
FR-2	Data Aggregation & Querying	State-wise aggregation queries Region-wise aggregation queries Time-based (yearly/monthly) aggregation queries
FR-3	Dashboard Visualization	Build Tableau dashboards with charts and maps Add interactive filters Build a narrative story from the dashboards
FR-4	Web Presentation	Serve dashboards/insights through a Flask web application Provide simple navigation between views

Non-functional Requirements

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboards and web pages are simple to navigate without training.
NFR-2	Security	Database access is restricted; no direct public access to SQL Server.
NFR-3	Reliability	Cleaned data pipeline produces consistent results on every run.
NFR-4	Performance	Aggregation queries and dashboard filters respond quickly on the dataset size used.
NFR-5	Availability	Web application is available whenever the Flask server is running.
NFR-6	Scalability	Pipeline stages are modular, allowing new datasets or years to

FR No.	Non-Functional Requirement	Description
		be added later.